

EE-323 Analog & Digital Communication

MID-I Exam

Roll #

Date: 19th September '16

Total Marks: 30

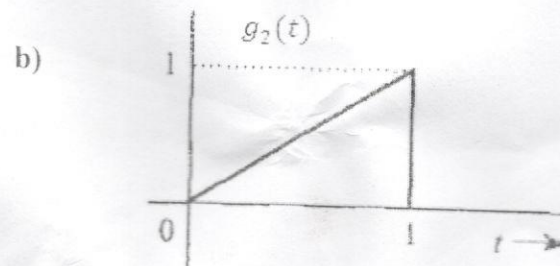
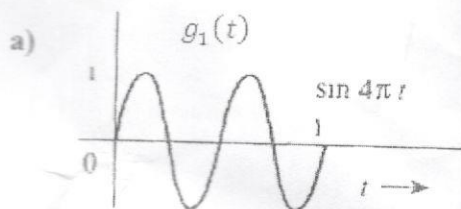
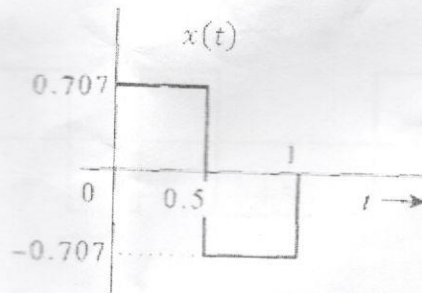
Time: 60 minutes

Total number of questions: 3. Fourier Transforms Table is provided.

Question No. 1

[10]

Find the correlation coefficient c_n of signal $x(t)$ and each of the two pulses $g_1(t)$ and $g_2(t)$ shown in figure below



Question No. 2

[10]

For the signal

$$x(t) = \frac{4}{t^2 + 4}$$

Determine the essential bandwidth B Hz of $x(t)$ such that the energy contained in the spectral components of $x(t)$ of frequencies below B Hz is 95% of the signal energy E_x .

$$B = \frac{\omega}{2\pi}$$

A random binary pulse train $g(t)$ is shown in figure below. The pulse width is T_w , and one binary digit is transmitted every T_w seconds. A binary 1 is transmitted by the positive pulse, and a binary 0 is transmitted by the negative pulse. The two symbols are equally likely and occur randomly.

1. Find the autocorrelation function $\mathcal{R}_g(\tau)$, plot it and give the rationale for computing it.
2. Find the power spectral density $S_g(\omega)$, plot it and comment on the spectral characteristics of the pulse train.

